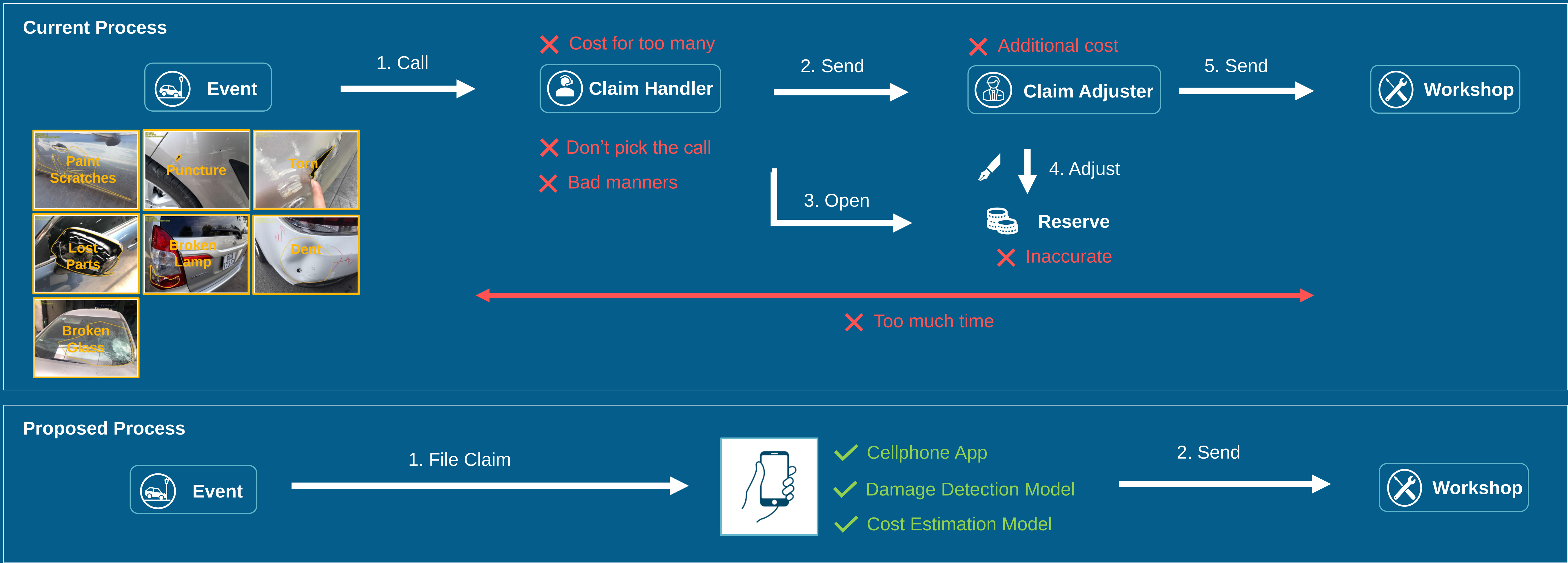


Insurance Claiming Efficiency: AI-Powered Car Damage Detection & Cost Automation



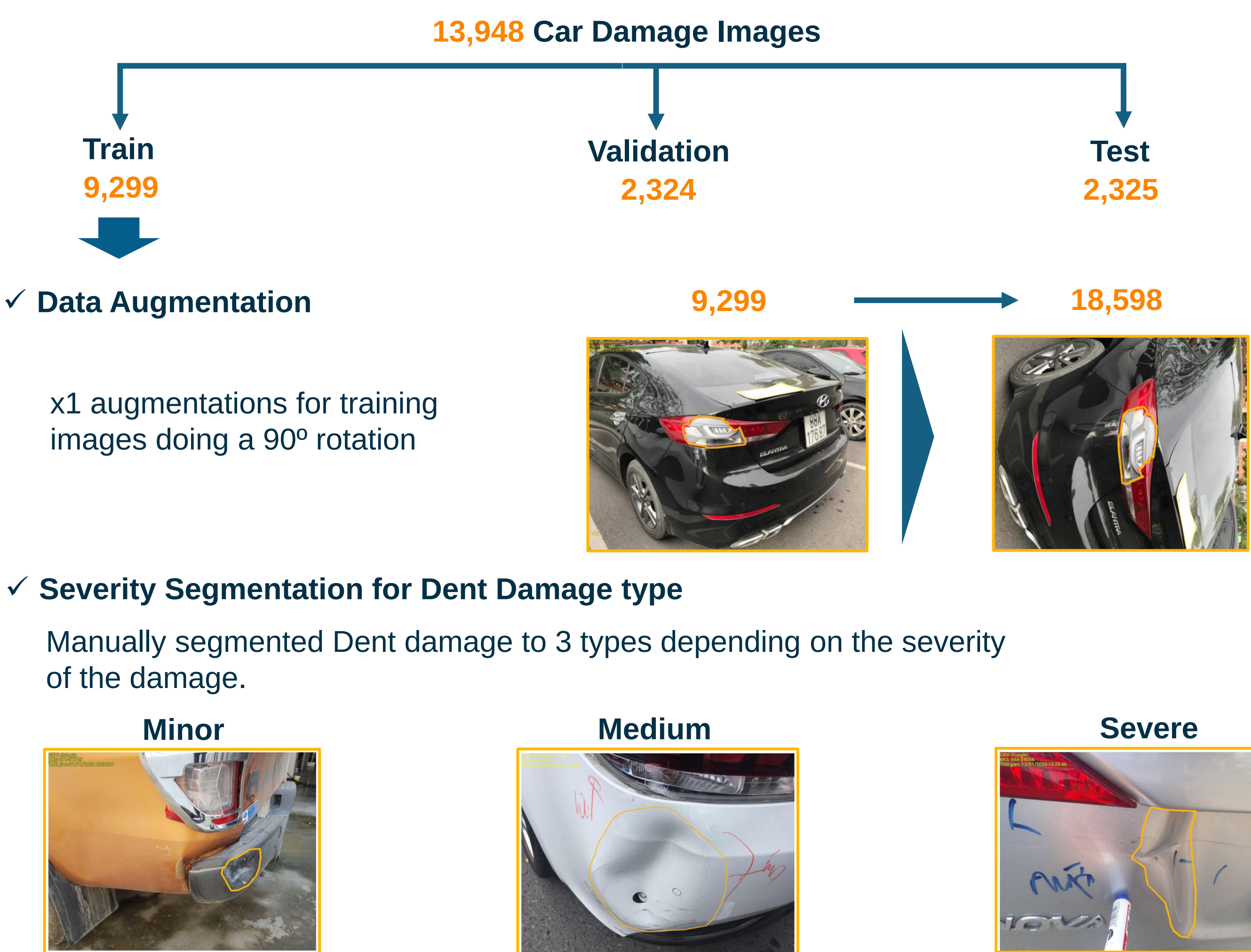
Teresa Babío Nuñez – 23201415 Daniel Quesada - 23201470

① Problem & Solution for High Frequency Claims Process



Damage Detection Model

Dataset



Modelling

Main Points

- ✓ Medium YOLOv8 model performing 100 epochs using Adam optimizer and early stopping with 30 patience and batch size of 32.
- ✓ Hyperparameters tuning:
 - Learning Rate: [0.0001, 0.001]
 - Momentum: [0.8, 0.9, 0.98]
 - Learning rate factor: [0.01, 0.1, 0.5, 1]
 - Weight Decay: [0.0001, 0.001, 0.01]
- ✓ Trained using AWS Sagemaker tool.



Modelling Results

Damage Type	MAP YOLO
All	40.4%
Lost Parts	66.7%
Torn	31.1%
Paint Scratches	26.4%
Broken Glass	77.9%
Puncture	50.0%
Broken Lamp	51.3%
Minor Dent	13.9%
Medium Dent	16.0%
Severe Dent	30.2%

Cost Estimation Model

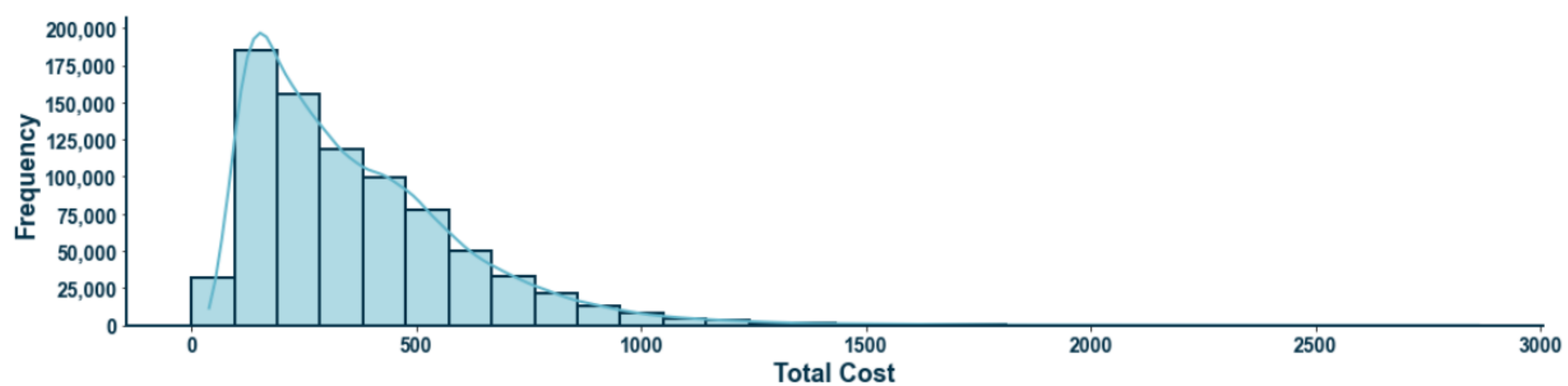
Dataset

- ✓ Synthetic claim dataset by damage type (9 different datasets).
- ✓ Each claim simulated in the following manner:

$$\text{Simulated Repair Cost} + \text{Simulated Labour cost by hour} \times \text{Simulated n}^\circ \text{ of hours}$$

- ✓ **Repair Cost:** Truncated gamma distribution by:
 - Damage type
 - Car producer
 - Model
 - Vehicle Age
- ✓ **Labour Cost by hour:** Fixed labour cost per hour modified by:
 - Workshop Quality
 - County
- ✓ **N° of hours:** truncated exponential distribution by damage type.

- ✓ 810,000 simulated claims (90,000 by damage type). The total cost distribution is the following:

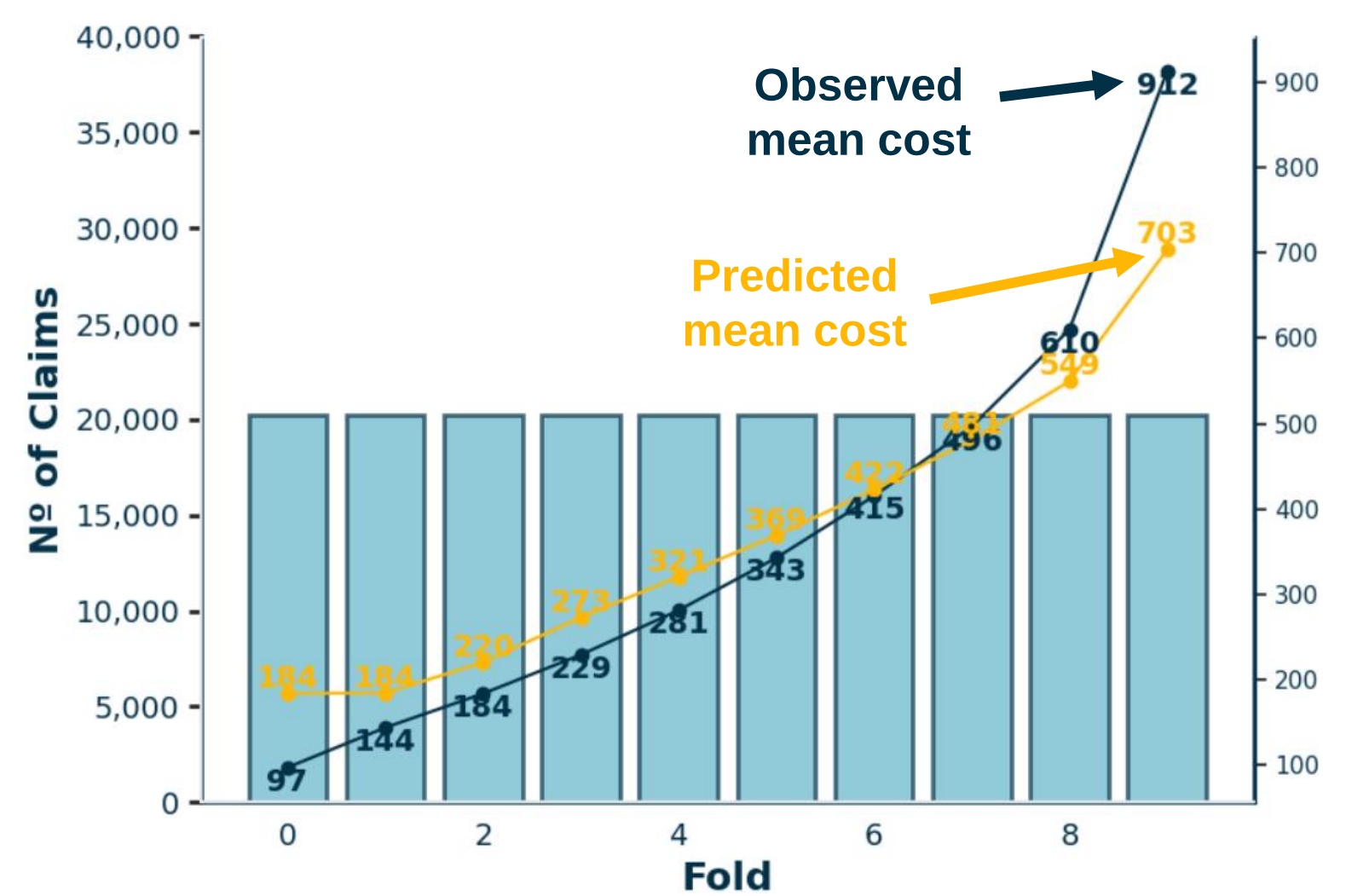


Modelling

Main Points

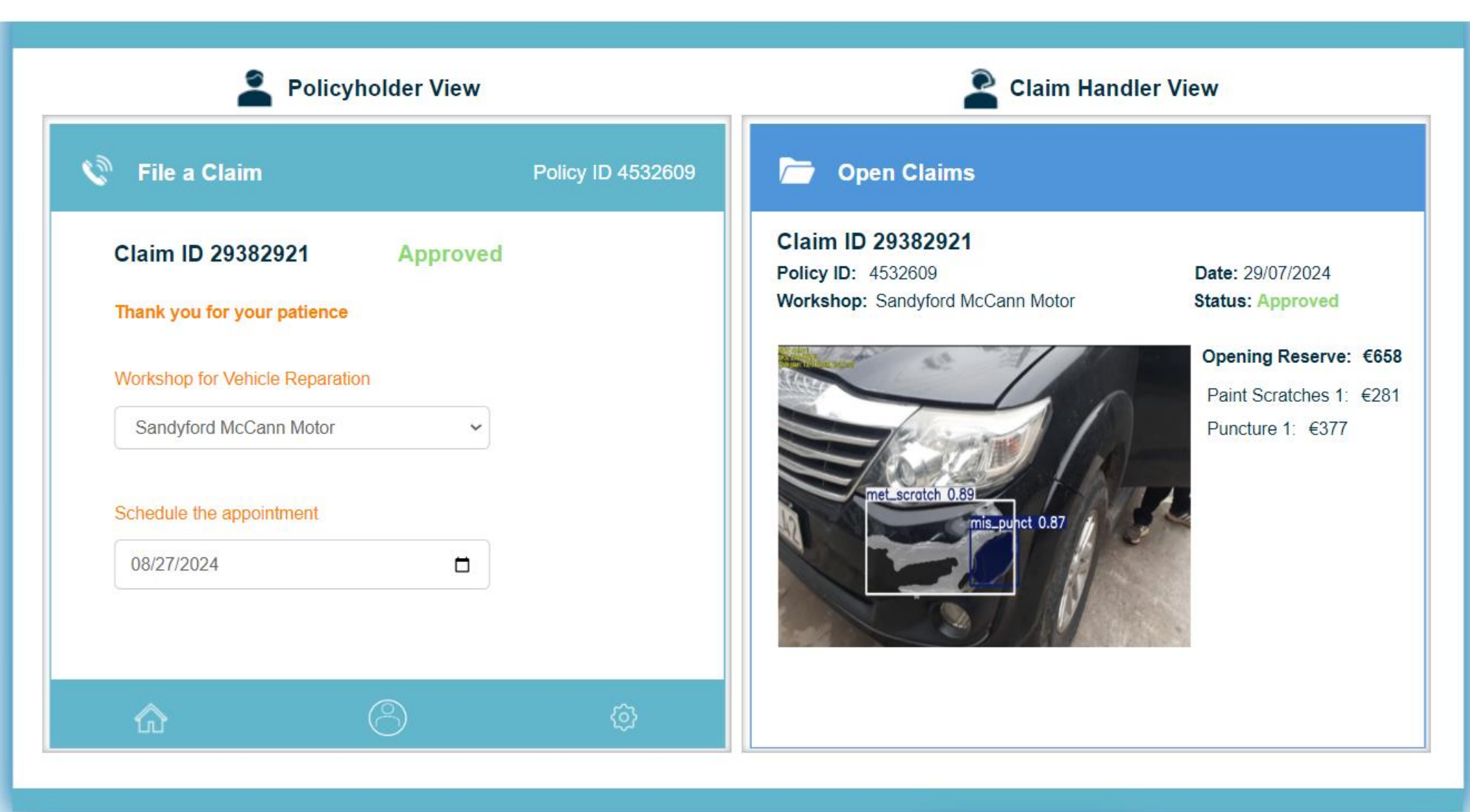
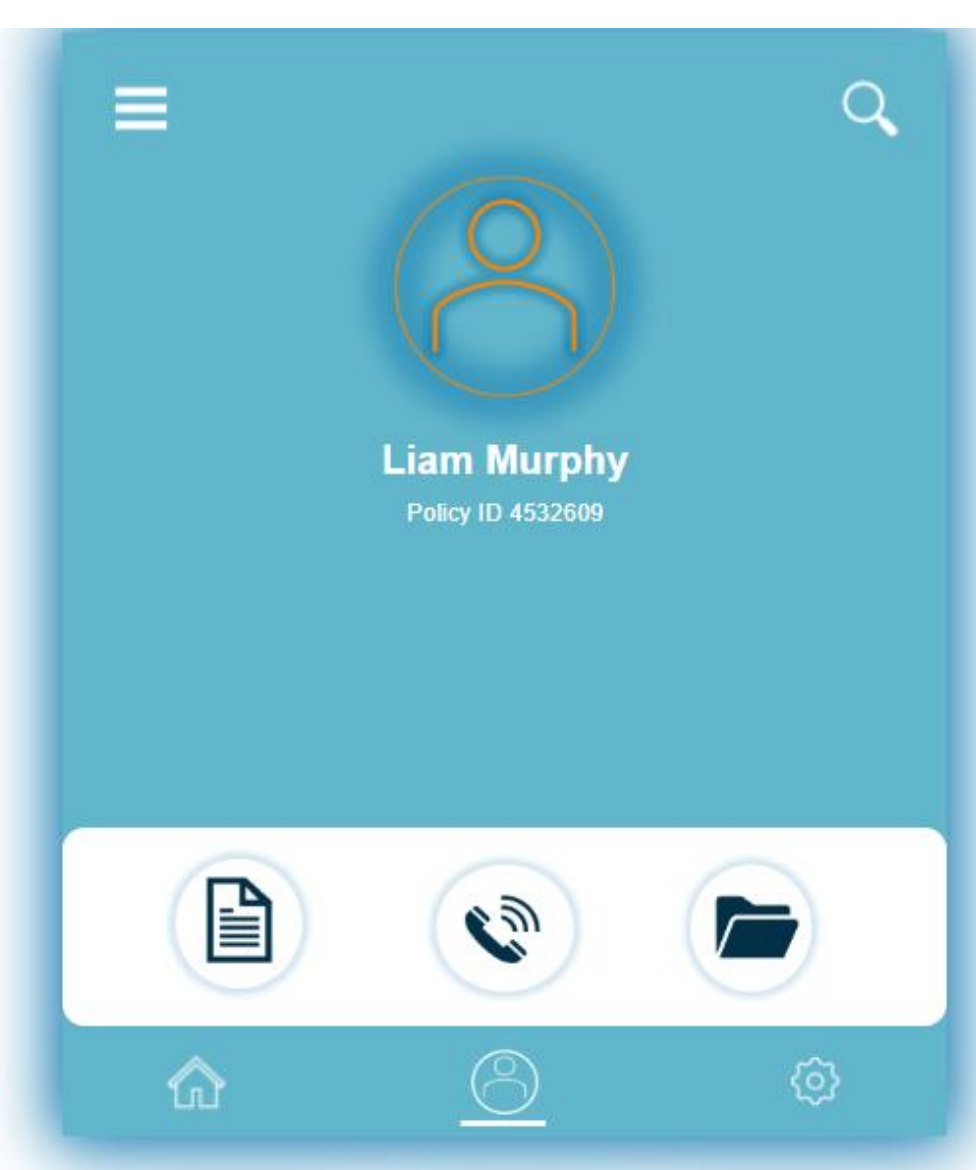
- ✓ Random Forest model.
- ✓ Using all the predictors to simulate the claim dataset.
- ✓ Model underfits, still better than initial estimate made for the insurance company.

Modelling Results



API

App simulation using Python Streamlit library



Future Work & References

New phases to enhance this project to be studied:

- ✓ Develop the API using a more suitable tool such as Flask.
- ✓ Get a real claim dataset and improve cost estimation model.
- ✓ Use a real claim adjuster to segment claim damage severity in pictures.

For accessing this project material and the references is based on:

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